

Tutor-Student Assignment Problem

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Problem Description

- **Objective:** Optimize the assignment of new students to available tutors.
- **Core Requirements:** Satisfy educational needs, honor tutor preferences, and respect operational capacities.
- **2 Scenarios:**

Scenario A: Resource Efficiency

Minimize the total active workforce while maximizing preference scores

Scenario B: Workload Equity

Equitably balance tutor load while maximizing tutor's preference on the tuition centre.

- **Tutor Mobility:** Tutors are mobile and can teach at multiple locations on different days. Existing assignments anchor capacity to a specific center, but unallocated capacity remains fluid across the organization.
- **Existing Assignment Priority:** To maintain tutor-student rapport and stability (especially during examination years), existing assignments consume capacity first and are not subject to re-optimization.

Parameters

- E_j : Number of existing students assigned to tutor j .
- Cap_j : Maximum overall capacity for tutor j .
- **Scenario Weights (α, β)**: Controls the trade-off between the scenario primary objective (efficiency in A or equity in B) and the fulfillment of tutor preferences.
- **Preference Weights ($P_{i,j}$)**:

$$P_{i,j} = \begin{cases} 2 & \text{if student's center matches tutor's 1st choice} \\ 1 & \text{if student's center matches tutor's 2nd choice} \\ 0 & \text{otherwise} \end{cases}$$

To solve the optimization, we define two types of discrete variables:

- **Assignment Variable ($x_{i,j}$):**
 - $x_{i,j} \in \{0, 1\}$
 - Binary assignment; 1 if student i is assigned to tutor j , 0 otherwise.
- **Tutor Active Variable (y_j):**
 - $y_j \in \{0, 1\}$
 - Binary assignment; 1 if tutor j is activated, 0 otherwise.
 - Used in Scenario A to indicate if tutor j is utilized.

Scenario A: Resource Efficiency

Objective: Minimize active tutors while maximizing preference scores.

$$\min Z_A = \alpha \sum_{j \in J} y_j - \beta \sum_{i \in I} \sum_{j \in J} P_{i,j} x_{i,j}$$

Subject to:

- **Assignment:** $\sum_{j \in J} x_{i,j} = 1 \quad \forall i \in I$
- **Skill Alignment:** $x_{i,j} = 0$ if Student is Extensive & Tutor is Normal
- **Net Capacity:** $E_j + \sum_{i \in I} x_{i,j} \leq Cap_j \quad \forall j \in J$
- **Utilization Logic:** $x_{i,j} \leq y_j \quad \forall i \in I, j \in J$

Scenario B: Minimax Balancing

Objective: Minimize peak workload across all tutors (L_{max}).

$$\min Z_B = \alpha L_{max} - \beta \sum_{i \in I} \sum_{j \in J} P_{i,j} x_{i,j}$$

Subject to:

- **Minimax Constraint:** $E_j + \sum_{i \in I} x_{i,j} \leq L_{max} \quad \forall j \in J$
- **Global Capacity:** $E_j + \sum_{i \in I} x_{i,j} \leq Cap_j \quad \forall j \in J$
- Skill Alignment and Assignment constraints apply as in Scenario A.

Code 1: Data Pre-processing & Parameters

- **Workload Filtering:** Extract only active existing students from historical data to determine current capacity (E_j).
- **Preference Scoring:** Generated a weighted matrix ($P_{i,j}$) where:
 - $C_i = P1_j \rightarrow 2$ points
 - $C_i = P2_j \rightarrow 1$ point
 - Otherwise $\rightarrow 0$ points
- **Integrity Checks:** Defined sets for 20 new students (I) and 10 tutors (J) to map binary decision variables.
- **Feasibility Checks:** Check if any tutors are overloaded with existing students already.

Code 2: 2 Scenario Optimization

Use docplex to solve two scenarios:

Scenario A: Efficiency

- **Goal:** Minimize active tutors.
- **Logic:** Uses activation variables (y_j) and Big-M constraints.

Scenario B: Equity

- **Goal:** Minimax workload.
- **Logic:** Minimizes the peak load variable (L_{max}).

Common Constraint: Skill alignment (Extensive-need students require Extensive-skilled tutors).

Code 3: Visual Comparison of Strategies

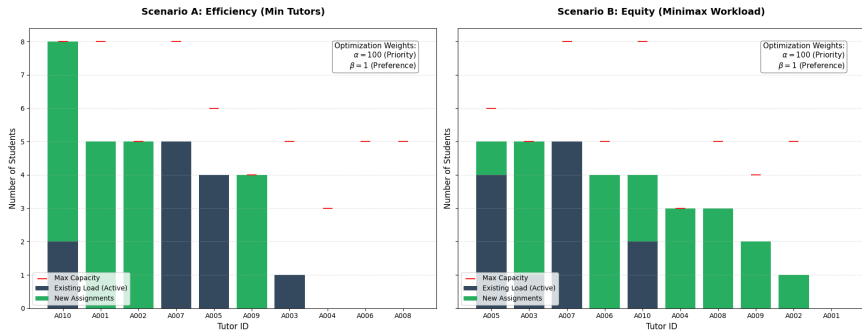


Figure: Scenario A: Efficiency vs. Scenario B: Equity

- **Scenario A:** High utilization per tutor, saving money but creates "spiky" workloads.
- **Scenario B:** Uniform load distribution, ensuring sustainable quality of service. Possibly higher cost or losing resources to competitors.

Decision Recommendation

Metric	Scenario A	Scenario B
Active Manpower	Minimal	Full Team
Individual Load	High (Spiky)	Balanced (Flat)
Preference Match	Lower	Higher (β dependent)
Operational Risk	High (Burnout)	Low (Resilient)

Conclusion and Further Extensions

Further model extensions within the domain of tuition assignment

- **Assignment in more dimensions** Include subjects (Physics, Geography, etc), discrete time-slots (day/time), and assignment levels (Sec 1-4, JC, etc).
- **Annual Re-optimization:** Allowing for a "complete shuffle" annually to re-align with the global optimum objectives while maintaining short-term stability when new students arrive on a rolling basis.
- **Penalizing Multi-location:** Tutors may not like traveling to too many locations.
Hard constraint- restrict to 2 location assignments
Soft constraint- add a penalty to third location.

End

Thank You